

Mathematics I

Week 2: Solve with us

Plan for this session

- How to join?
 - Join on webex/zoom - click on link sent to you
 - Join on pear deck - joinpd.com (enter code: see top right on webex/zoom)
 - Keep a notebook and pen ready for solving problems
- For every question - 5 to 15 minutes allotted
 - Question will be shown in a slide for solving
 - If you are done solving, enter your answer on joinpd.com
 - Presenter will provide a solution
 - Questions and discussion

Sample Question - your screen on joinpd.com

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Description of the
problem.

Question to be
answered.

Desktop

Answer box

enter a number or
a choice or some
text

Mobile

Answer question

Example Screenshots

Laptop/Desktop

The screenshot shows a presentation slide with a dark background. On the left, there is a large grey rectangular area labeled "Q1 (a)". To its right, the text reads: "Is this function even or odd or neither even nor odd?". In the top right corner, a small box contains the text "How to participate? joinpd.com code". In the bottom left corner, there is a small cartoon character and the text "Students choose an option". In the bottom right corner, it says "Pear Deck Interactive Slide Do not remove this bar".

Portion for Answering

Even

Odd

Neither even nor odd

Mobile

The screenshot shows the same presentation slide on a mobile phone screen. The top status bar shows the time as 2:50 PM and various icons. The address bar shows "app.peardeck.com/studi". The slide content is the same as the laptop/desktop version. A red circle highlights the "Answer Question" button in the bottom right corner. An arrow points from the text "Portion for Answering" to this button.

Portion for Answering

Answer Question

Test Problem

This is a problem to test if everything is working.

Answer the following:

- a. I am working professional
- b. I am college/school student



Students choose an option

Q1

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Points $P(-1, -2)$, $Q(1, 2)$ and $R(g, h)$ are collinear.

Hint:

Collinear points: Three or more points that lie on a same straight line.

Identify the coordinates of the point R from the following:

- a) $(2, 4)$
- b) $(3, -1)$
- c) $(1, 3)$
- d) $(-2, -4)$



Students, write your response!

Solution:

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Step 1:

- Since the points 'P', 'Q' and 'R' are collinear then their slope must be equal.
- That is slope of 'PQ' = slope of 'QR'
- Formula for slope = $(y_2 - y_1) / (x_2 - x_1)$
 - $\Rightarrow (2 - (-2)) / (1 - (-1)) = (h - 2)/(g - 1)$
 - $\Rightarrow h = 2g$

Step 2:

- Clearly, twice the X-coordinate of 'R' is Y-coordinate
- So, **option (a)** and **option (d)** are right

Q2

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A publishing house purchases a printing machine for Rs. 50,000. At the end of 5 years the value of the machine is supposed to be Rs. 10,000 only.

If the loss in value is assumed to be linear then what is the yearly loss in the value of machine?

- a) Rs. 5,000 only
- b) Rs. 6,000 only
- c) Rs. 10,000 only
- d) Rs. 8,000 only



Students choose an option

Solution:

Step 1:

- Let 'y' represent the value of the machine and 'x' represent the number of years after purchasing the machine.

Step 2:

- $x = 0 \Rightarrow y = \text{Rs. } 50,000$ (purchased value of the machine)
- $x = 5 \Rightarrow y = \text{Rs. } 10,000$ (Value of the machine at the end of 5 years)

Step 3:

- What will be the yearly loss of the value of machine?
 - **Slope** will give us the yearly loss of the value of machine
 - $\text{Slope} = (y_2 - y_1) / (x_2 - x_1) = (10000 - 50000) / (5 - 0) = -8000$
- Therefore yearly loss in the value of the machine is **Rs. 8000**

Solution:

Step 1:

- Let 'y' represent the value of machine and 'x' represent the number of years after purchasing the machine.

Step 2:

- When $x = 0 \Rightarrow y = \text{Rs. } 50,000$ (purchased value of the machine)
- When $x = 5 \Rightarrow y = \text{Rs. } 10,000$ (Value of the machine at the end of 5 years)

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- Therefore yearly loss in the value of the machine is **Rs. 8000**

Q: What will be the equation to find the value of machine for any particular year?

Solution:

Step 1:

- Let 'y' represent the value of machine and 'x' represent the number of years after purchasing the machine.

Step 2:

- When $x = 0 \Rightarrow y = \text{Rs. } 50,000$
- When $x = 5 \Rightarrow y = \text{Rs. } 10,000$

Step 3:

- What will be the yearly loss of the value of machine?
 - **Slope** will give us the yearly loss of the value of machine
 - $\text{Slope} = (y_2 - y_1) / (x_2 - x_1) = (10000 - 50000) / (5 - 0) = -8000$
- Therefore yearly loss in the value of the machine is **Rs. 8000**

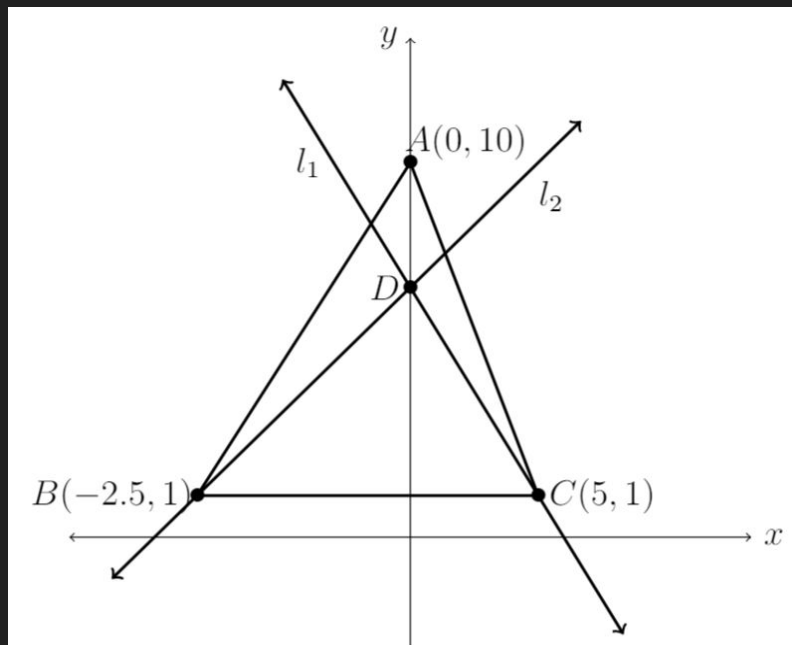
Q: What will be the equation to find the value of machine for any particular year?

Answer: $y = 50000 - 8000x$

Q3.1

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The coordinates of points A , B , and C are shown in Figure. Suppose the slope of the line l_2 is 2 and D is the point of intersection of the two lines l_1 and l_2 .



Which of the following statements are correct?

- a) The point D will be at $(0, 5)$.
- b) The equation of the line l_1 is $x + y = 6$.
- c) The line l_1 is perpendicular to the line l_2 .
- d) The line l_1 is parallel to the line l_2 .



Students choose an option

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code: see above

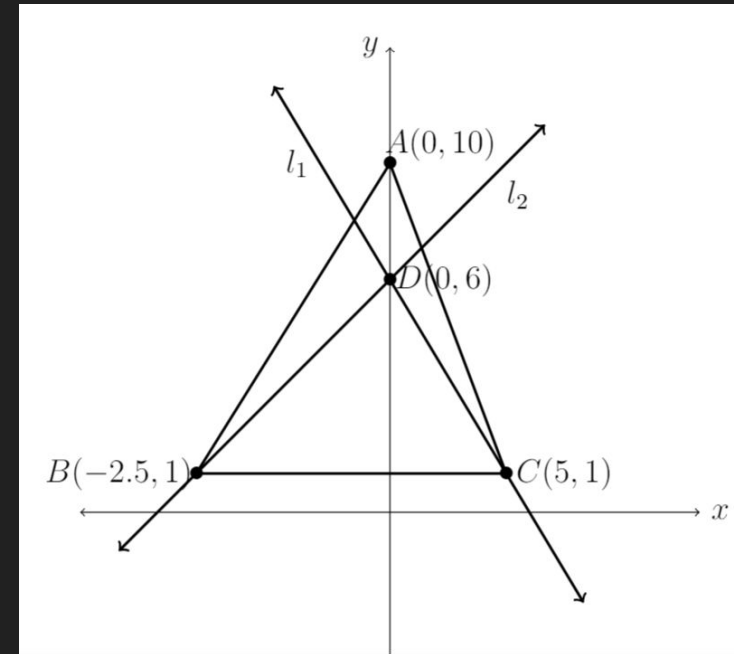
Solution:

Step 1: Finding equation of line l_2

- Given that slope (m_2) of the line l_2 is 2 and it passes through $(-2.5, 1)$
- Find the equation of line using point slope form
 - $y = 2x + 6$ (l_2)

Step 2: Finding the coordinates of D

- X-coordinate of D is 0 (given in figure).
- We know that line l_2 passes through D
 - $y = 2(0) + 6 \Rightarrow y = 6$
 - Thus, point $D(0, 6)$



How to participate?
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Solution:

Step 3: Finding equation of line l_1

- Given that line l_1 passes through point $D(0, 6)$ and $C(5, 1)$
- Equation of the line l_1 can be found using two point form
 - $y = -x + 6$ (l_1)
 - Slope (m_2) of the line l_1 is -1

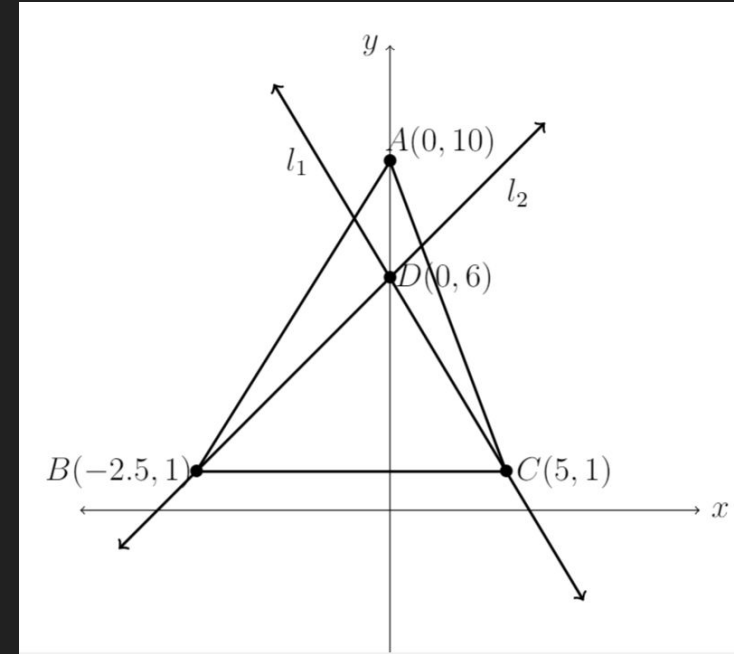
Step 4: Check if they are perpendicular

- If $m_1 \times m_2 = -1$ (then perpendicular)
- $2 \times -1 = -2 \neq -1$ (thus not perpendicular)

Step 5: Check if they are parallel

- If $m_1 = m_2$ (then parallel)
- $-1 \neq 2$ (thus not parallel)

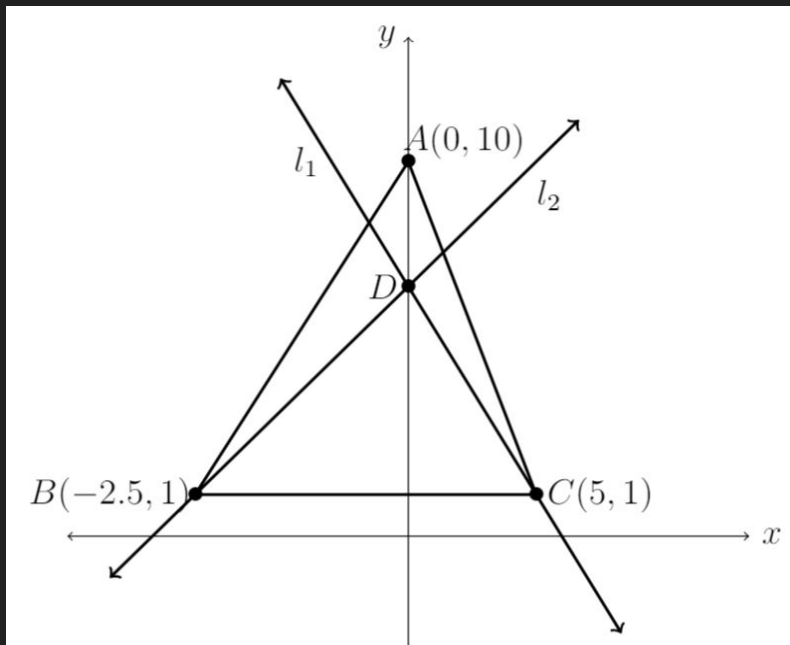
Thus, only options (b) is correct



Q3.2

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The coordinates of points A , B , and C are shown in Figure. Suppose the slope of the line l_2 is 2 and D is the point of intersection of the two lines l_1 and l_2 .



Suppose the ratio of the area of triangle ABC to the area of triangle DBC is $k : 1$. Find the value of k .



Students, enter a number!

Solution:

Step 1: Finding area of triangle

- Area of triangle = $\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$
- Area of triangle ABC where $A(0, 10)$, $B(-2.5, 1)$ and $C(5, 1)$
 - $= \frac{1}{2} |0(1-1) - 2.5(1-10) + 5(10-1)| = 33.75$
- Similarly, area of triangle DBC will be 18.75

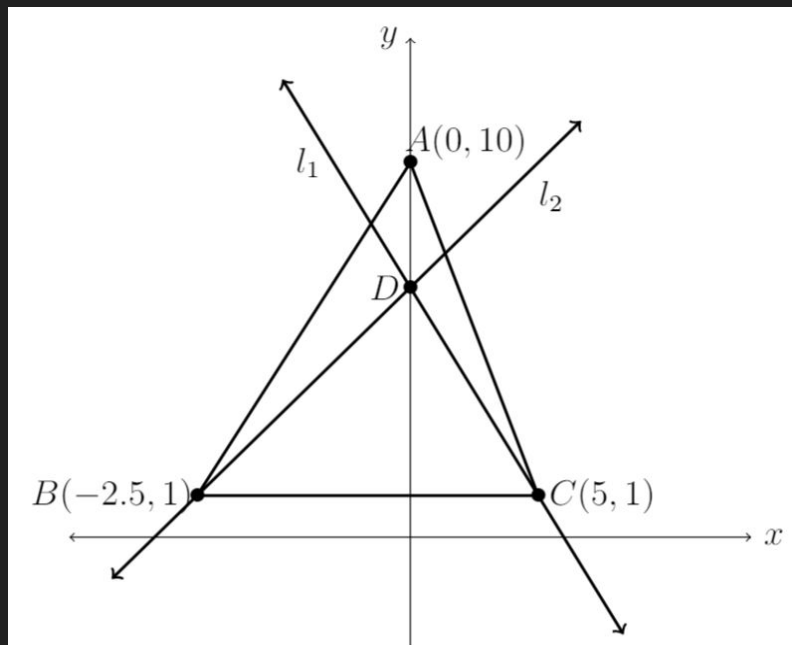
Step 2: Finding the value of k

- Area of triangle $ABC/DBC = k/1$
 - $33.75/18.75 = k/1$
 - $k = 1.8$

Q3.3

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The coordinates of points A , B , and C are shown in Figure. Suppose the slope of the line l_2 is 2 and D is the point of intersection of the two lines l_1 and l_2 .



Consider a new line l_3 which passes through the point A and is parallel to the X -axis. What will be the area of the region enclosed by the lines l_1 , l_2 , and l_3 .



Students, enter a number!

Solution:

How to participate?
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Step 1: Find the equation of the line l_3

- Equation of the line passing through point A and is parallel to X -axis (slope is 0) will be
 - $y=10$

Step 2: Find the intersection point (P) of lines l_1 and l_3

- Substitute and solve, then, we get coordinates of P as $(-4, 10)$

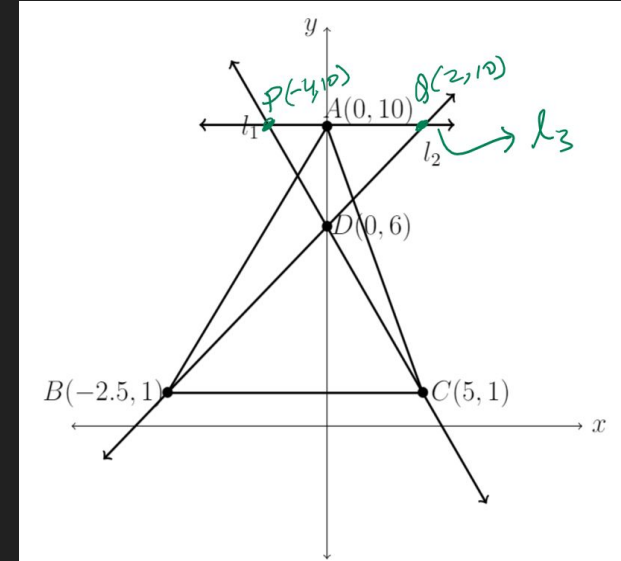
Step 3: Find the intersection point (Q) of lines l_2 and l_3

- Substitute and solve, then, we get coordinates of Q as $(2, 10)$

Step 4: Find the area of triangle DPQ

(i.e., area enclosed by l_1 , l_2 and l_3)

- On solving we get **12**



Q4

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A line is represented by ($7y + 8x = 56$). If mirror image is taken with respect to the Y-axis, a new line is formed.

Hint: Only the sign of the X-coordinate changes when Y-axis is mirror.

What will be the equation of the new line?

- a) $7x + 8y = 56$
- b) $y = 8/7x + 8$
- c) $y = 7/8x - 7$
- d) $x/8 + y/7 = 1$



Students choose an option

Solution:

Step 1:

- Rearranging the equation ($7y + 8x = 56$) in intercept form ($y/a + x/b = 1$) where 'a' and 'b' are Y and X intercepts respectively.
- Divide the above equation by 56 we get $y/8 + x/7 = 1$
- So, Y-intercept is 8, X-intercept is 7.

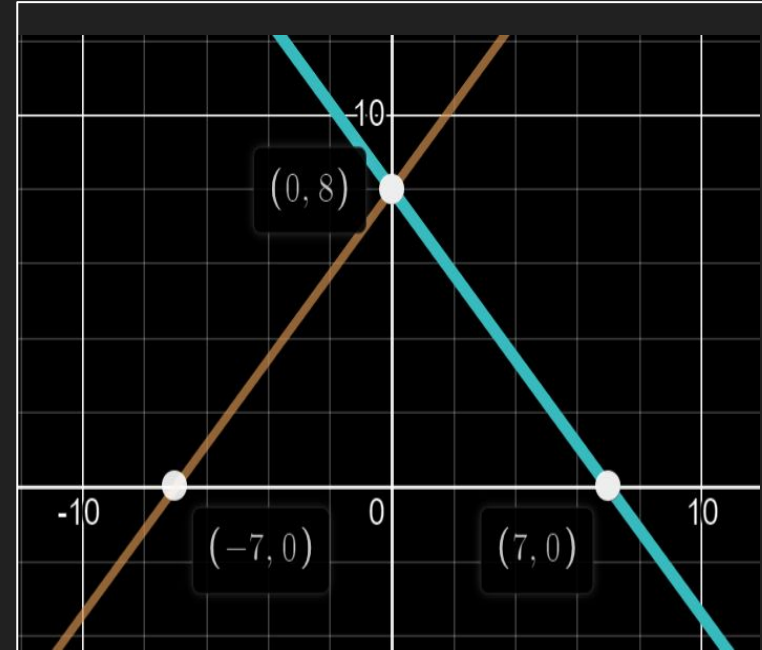
Step 2:

- Only the sign of the X-coordinate changes when Y-axis is mirror. (See Figure)
- Thus, the new X-intercept is (-7)

Step 3:

- Using intercepts form to get new equation of line $y/8 + x/(-7) = 1 \Rightarrow y = 8/7x + 8$

How to participate?
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code: see above



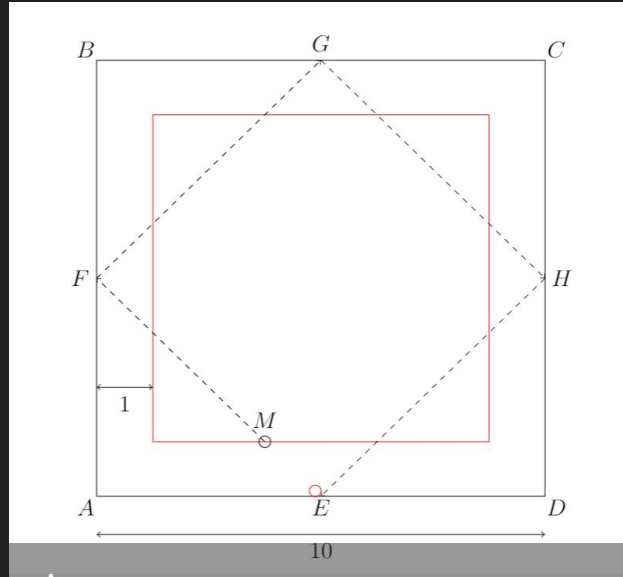
Desmos plot:

Blue color: $7y + 8x = 56$

Brown color: $y = 8/7x + 8$

Q5

A carrom board is a square board with a symmetrical design as shown in the diagram below. 'E', 'F', 'G', and 'H' are the midpoints of 'AD', 'BA', 'CB', and 'DC' respectively. Rohit wants to pocket his last coin which is at 'E'. Taking the laws of reflection to be applicable, he strikes from the point 'M', so that the striker reflects at 'F', then at 'G', then at 'H' and finally hits the coin at 'E'.



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Choosing 'A' to be the origin, Which of the following options is (are) correct.

- a) 'FH' and 'EG' are perpendicular line segments.
- b) Segment 'GH' intersects the inner square at (6, 9) and (9, 6).
- c) Slope of line 'EF' is -1.
- d) Distance between 'M' and 'E' is $\sqrt{2}$.

Students, write your response!

Solution:

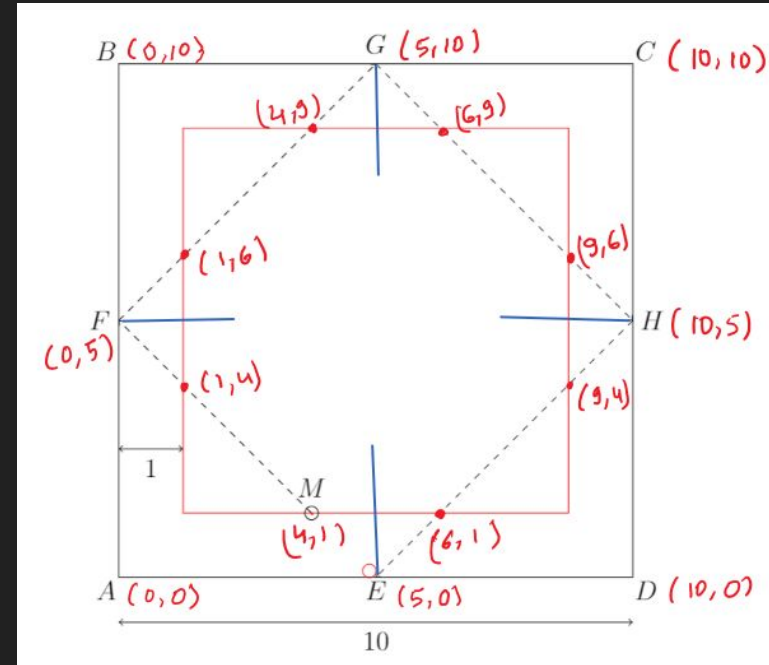
Step 1:

- Recall for square all sides are equal.
- So, using the properties of square, coordinates of 'A', 'B', 'C', 'D' is shown in Figure.

Step 2:

- Since 'E', 'F', 'G' and 'H' are midpoints 'AD', 'AB', 'BC' and 'CD' respectively. The coordinates will be as shown in Figure.
 - Formula: X-coordinate = $(mx_2 + nx_1)/2$; where $m = n = 1$;
 $\Rightarrow (x_2 + x_1)/2$
 - Formula: Y-coordinate = $(y_2 + y_1)/2$

How to participate?
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code: see above



Solution:

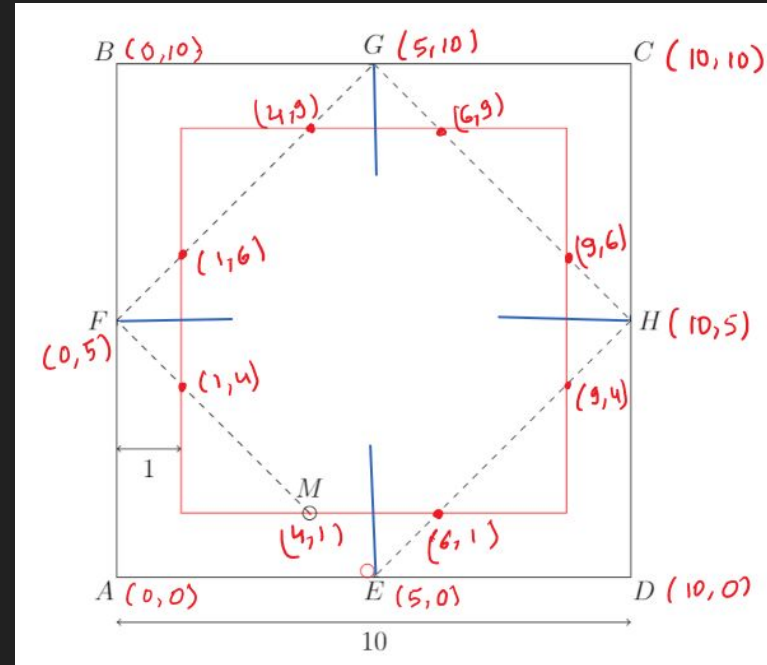
Step 3: To find intersection point of inner square by line segment 'GH'

- First find the equation of line segment 'GH'
- This can be done using two point formula.
 - $y = -x + 15$
 - So for $y = 9$; we get $x = 6$
and for $x = 9$; we get $y = 6$
 - Thus $(6, 9)$ and $(9, 6)$ are the intersection points as shown in Figure.

Step 4:

- 'FH' and 'EG' are perpendicular line segments.
- Use property $(m_1 \times m_2) = -1$

How to participate?
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code: see above



Solution:

Step 5: Based on the symmetrical design of Carrom board, the coordinates of all the intersection points of the inner square can be found.

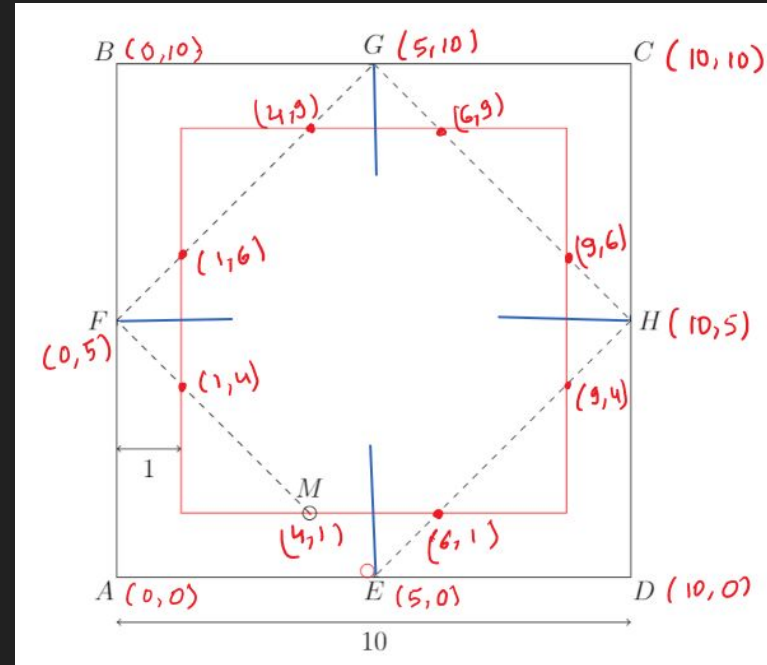
Step 6: Distance between point M and E

- The distance of 'ME' can be calculated using distance formula. And the distance is found to be $\sqrt{2}$
- Note: Distance formula is

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

All options are correct

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Q6

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code: see above

A line perpendicular to the line segment joining the points A(1, 0) and B(2, 3), divides it at C in the ratio of 3 : 1.

Then the equation of perpendicular line (in general form) is:

- a) $6x + 3y - 17 = 0$
- b) $y = -\frac{1}{3}x + 17$
- c) $2x + 6y - 17 = 0$
- d) $4x + 12y - 17 = 0$



Students choose an option

Solution:

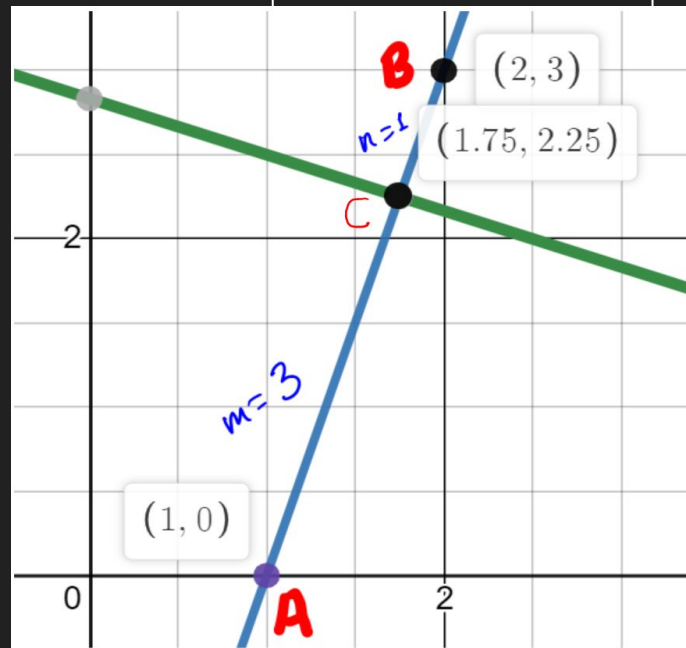
How to participate?
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Step 1: Finding the slope of perpendicular line

- Slope of line segment 'AB' is $(3 - 0)/(2 - 1) = 3$ (say m_1)
- Slope of Perpendicular line (m_2) is
- $m_2 = -1/m_1 \Rightarrow m_2 = -1/3$

Step 2: Finding the coordinates of intersection point (C)

- $C = \left(\frac{m \times x_2 + n \times x_1}{m + n}, \frac{m \times y_2 + n \times y_1}{m + n} \right)$
- $C = \left(\frac{3 \times 2 + 1 \times 1}{4 + 1}, \frac{3 \times 3 + 1 \times 0}{4 + 1} \right)$
- $C = (7/4, 9/4)$



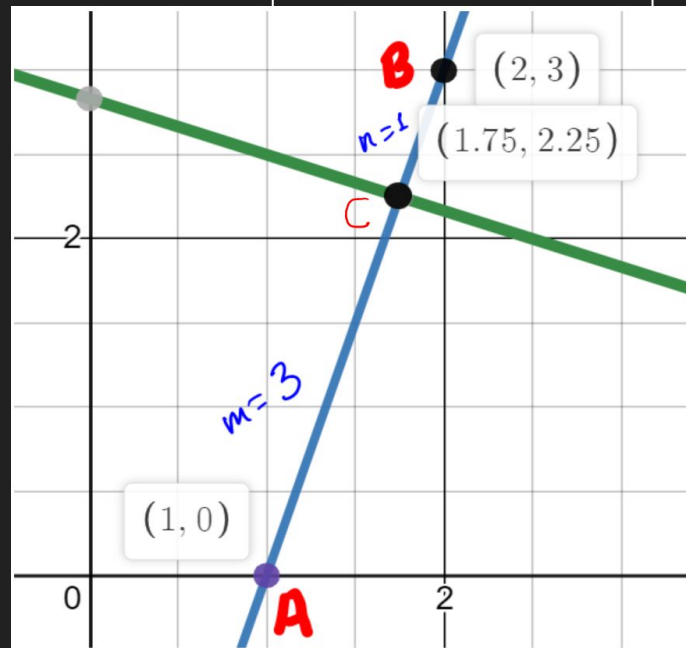
Desmos plot:
Green line is perpendicular line

Solution:

Step 3: Finding the equation of line

- We have slope of perpendicular line i.e., $m_2 = -1/3$ and it passes through $C = (7/4, 9/4)$
- Equation of line in slope point form will be
 - $(y - y_0) = m(x - x_0)$
 - $(y - 9/4) = -1/3 (x - 9/4)$
 - $2x + 6y - 17 = 0$ (this is general form of straight line)

How to participate?
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code: see above



Desmos plot:
Green line is perpendicular line

SSE

How to participate?
joinpd.com
code: see above

Q1. What is SSE and what is it used for?

Answer:

SSE

How to participate?
joinpd.com
code: see above

Q1. What is SSE and what is it used for?

Answer:

- SSE is Squared Sum Error
- It is calculated using SSE formula $= \sum_{i=1}^n (y_i - (mx_i + c))^2 = \sum_{i=1}^n (y_i - mx_i - c)^2$
- Where y_i is the actual value and $(mx_i + c)$ is the predicted values
- It can be used to calculate the best fit for the given data set.

SSE

How to participate?
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code: see above

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Q2. Can SSE be negative?

Answer:



Students choose an option

SSE

How to participate?
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code: see above

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Q2. Can SSE be negative?

Answer:

No, because $SSE = \sum_{i=1}^n (y_i - (mx_i + c))^2$ is the squared function. So the least possible value is 0

SSE

How to participate?
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code: see above

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No, because $SSE = \sum_{i=1}^n (y_i - (mx_i + c))^2$ is the squared function. So the least possible value is 0

Q3. Given below are three SSE values which among the following values will depict the best fit?

(a) 4 (b) 5 (c) 8

Students choose an option



SSE

How to participate?
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code: see above

Q3. When do we say best fit in general?

Answer:

SSE

How to participate?
joinpd.com
code: see above

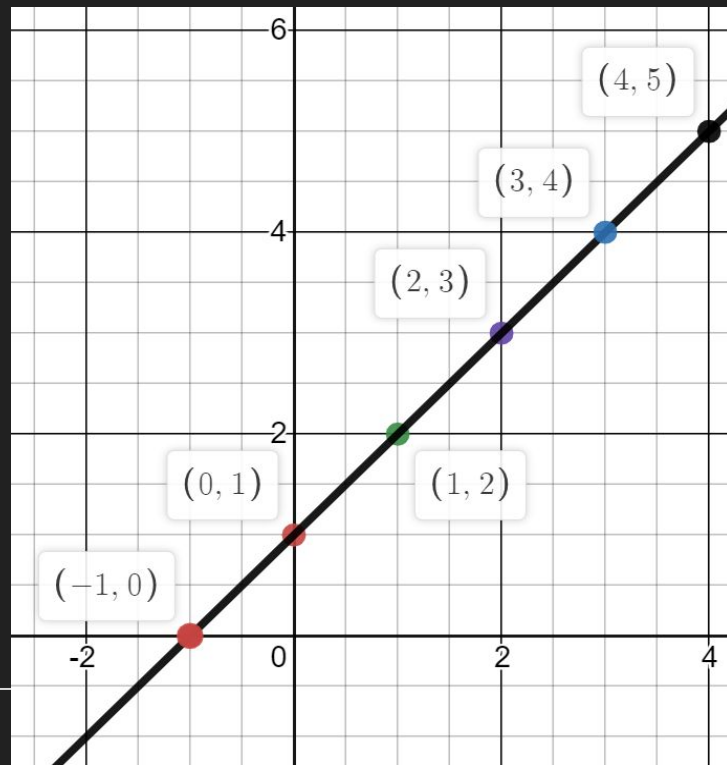
Q3. When do we say best fit in general?

Answer:

- When predicted value = actual value i.e., SSE is 0

Note:

If asked to choose best fit value from the given options choose the least value if it fulfils the given conditions.



Q7

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An experiment is performed in an ideal condition to find out the velocity of a particle obeying the equation ($S = vt$) where the distance travelled (S) inside a labeled glass tube by a particle with an uniform velocity (v) is measured with respect to time taken (t). The data collected by repeating the experiment five times is shown in the table below.

S (in cm)	3	9	13	14	17
t (in sec)	2	4	6	8	10

Hint : Use the option values as ' v ' in ($S = vt$) and calculate the SSE

If we use the method to minimize the squared sum error, what is the most likely velocity of the particle among the given options?

- a) 2.5
- b) 1.5
- c) 2
- d) 1



Students, write your response!

Solution:

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S (in cm)	t (in sec)	$(S - vt)^2$			
		$v = 2.5$	$v = 1.5$	$v = 2$	$v = 1$
3	2	4	0	1	1
9	4	1	9	1	25
13	6	4	16	1	49
14	8	36	4	4	36
17	10	64	4	9	49
SSE		$\Sigma = 109$	$\Sigma = 33$	$\Sigma = 16$	$\Sigma = 160$

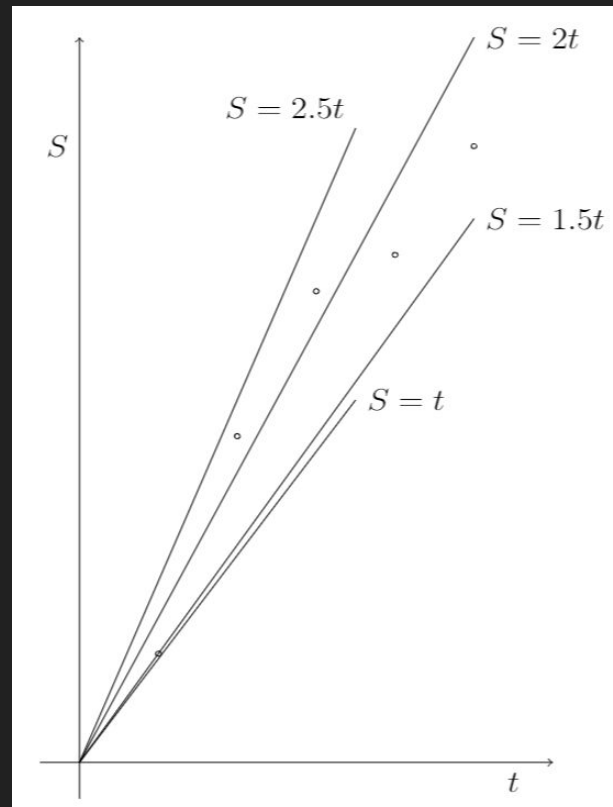
- From the table, we see that ($v = 2$) gives the minimum Sum Square Error.
- Most likely velocity of the particle is 2.

Solution:

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S (in cm)	t (in cm)	$(S - vt)^2$			
		$v = 2.5$	$v = 1.5$	$v = 2$	$v = 1$
3	2	4	0	1	1
9	4	1	9	1	25
13	6	4	16	1	49
14	8	36	4	4	36
17	10	64	4	9	49
SSE		$\Sigma = 109$	$\Sigma = 33$	$\Sigma = 16$	$\Sigma = 160$

- From the table, we see that ($v = 2$) gives the minimum Sum Square Error.
- Most likely velocity of the particle is 2.



Q8

Mrs. Ruby is a researcher, she conducts research on the decay of the quality of bananas. She proposed 5 models:

$$M_1 : y = x + 20$$

$$M_2 : y = 1.5x + 20$$

$$M_3 : y = -2x + 15$$

$$M_4 : y = -4x + 18$$

$$M_5 : y = -3x + 20$$

In these models, y indicates a quality factor (or decay factor) which is dependent on x number of days. The value of y varies between 0 and 20, where the value 20 denotes that the fruit has no decay and $y = 0$ means it is completely decayed. While formulating a model she has to make sure that on the 0 th day the banana has no decay. The quality factor (or decay factor) y values on x th day are shown in Table 1

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code: see above

What is the SSE value of the best fit model till two decimal places?

x	y
4	6
5	4
6	3
7	1

Table 1

Hint: SSE formula

$$\sum_{i=1}^n (y_{\text{given value}} - y_{\text{predicted value}})^2$$

Students, enter a number!

Solution:

Step 1: Identify the right model

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- Note M_3 and M_4 cannot be considered as the right model as she has to make sure that 0 th day (i.e., when $x = 0$) the value of $y = 20$
- Also note M_1 and M_2 models can also be ignored as the SSE value for these model will be too large because these the slope is positive and data given in the table is of decreasing type (or decaying)
- So the only model which can be used for calculating SSE will be M_5

Solution:

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Step 2: Calculate SSE using M_5

- $$SSE = \sum_{i=1}^n (y_i - mx_i - c)^2$$

- | x | y | $(y - (-3x + 20))^2$ |
|---|---|----------------------|
| 4 | 6 | 4 |
| 5 | 4 | 1 |
| 6 | 3 | 1 |
| 7 | 1 | 4 |

Answer: 10

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code: see above

THANK YOU